Final revision paper

CAPS Grade 9 Mathematics · Terms 1 to 4

This is an original practice paper. It follows the CAPS topics and the 2025 Grade 9 Annual Teaching Plan, and it uses the same topic names as Platinum Mathematics Grade 9. It is not a copy of the textbook and it is not a past exam.

Time	2 hours for Questions 1 to 11	**Marks**	100, plus an optional Question 12
Calculator	Non-programmable calculator allowed	**Name**	_____

Instructions

1. Answer **Questions 1 to 11**. Together they are worth 100 marks.
2. Question 12 is extra practice from the data and probability chapters in Platinum. Do it if your teacher included those chapters. It is not part of the 100 marks and it is not part of the 2 hours.
3. Show **all** working. If a final answer is wrong, method marks can still be earned.
4. In geometry, give a reason for every statement.
5. Give answers in the units asked for. Write money in rands and cents.
6. Diagrams are not drawn to scale.
7. Round only when the question asks you to round.

QUESTION 1 · Whole numbers · 8 marks

1.1 Choose the **most precise** description for each number. Write only the letter. **(4)**

A natural number
B whole number, but not a natural number
C integer, but not a whole number
D rational, but not an integer
E irrational

(a) 0 (b) −11 (c) 2/9 (d) $\sqrt{7}$

1.2 Write 90 as a product of its prime factors, in exponential form. **(2)**

1.3 Find the HCF and the LCM of 28 and 42. **(2)**

QUESTION 2 · Ratio, rate, proportion and finance · 10 marks

2.1 A bus travels 210 km in 2 hours 30 minutes. Calculate its average speed in km/h. **(2)**

2.2 The ratio of red beads to blue beads is 5 : 3. There are 48 beads altogether. How many red beads are there? **(2)**

2.3 4 m of fabric costs R78. Calculate the cost of 7 m of the same fabric. **(2)**

2.4 It takes 6 machines 8 hours to print a batch of books. How long will 4 machines take if they work at the same rate? **(2)**

2.5 R4 000 is invested at 5% per annum, compounded yearly, for 2 years. Calculate the value of the investment after 2 years. **(2)**

QUESTION 3 · Integers · 6 marks

3.1 Calculate. **(6)**

3.1.1 $-8 + (-6)$ (1)
3.1.2 $4 - (-7)$ (1)
3.1.3 $(-3) \times (-4) \times (-2)$ (2)
3.1.4 $-36 \div 4 + (-2)^2$ (2)

QUESTION 4 · Exponents · 10 marks

4.1 Simplify. Where a power is negative, write the answer with a positive exponent. **(9)**

4.1.1 $y^6 \times y^2$ (1)
4.1.2 $(5^2)^2$ Write the answer as a number. (1)
4.1.3 $(2p^2q)^3$ (2)
4.1.4 $3^{-2} + 7^0$ (2)
4.1.5 $(8x^6y^2) \div (2x^2y^5)$ (3)

4.2 Write 0,0064 in scientific notation. **(1)**

QUESTION 5 · Algebraic expressions · 12 marks

5.1 The expression $-5x^2 + x - 9$ is given. **(2)**

5.1.1 Write down the coefficient of x^2. (1)
5.1.2 Write down the constant term. (1)

5.2 Simplify. **(6)**

5.2.1 $6a - 4b - a + 7b$ (1)
5.2.2 $5(x - 2) - 3(2x + 1)$ (3)
5.2.3 $(18m^2n - 12mn^2) \div 6mn$ (2)

5.3 Expand and simplify. **(4)**

5.3.1 $(x + 7)(x - 3)$ (2)
5.3.2 $(2x - 3)^2$ (2)

QUESTION 6 · Factorising and algebraic fractions · 10 marks

6.1 Factorise fully. **(8)**

6.1.1 $7x^2 - 21x$ (2)
6.1.2 $16 - p^2$ (2)
6.1.3 $x^2 + 2x - 15$ (2)
6.1.4 $2x^2 - 8x - 10$ (2)

6.2 Simplify $(x^2 - 16) / (x + 4)$. **(2)**

QUESTION 7 · Equations · 10 marks

7.1 Solve for x. **(8)**

7.1.1 $4x + 3 = -13$ (2)
7.1.2 $2(x - 5) = 3x + 1$ (3)
7.1.3 $x^2 - 9x = 0$ (3)

7.2 A mother is three times as old as her son. The sum of their ages is 48 years. How old is the son? **(2)**

QUESTION 8 · Patterns, functions and graphs · 10 marks

8.1 The pattern is 8 ; 13 ; 18 ; 23 ; … **(4)**

8.1.1 Write a formula for the general term, T_n. (2)

8.1.2 Which term is equal to 78? (2)

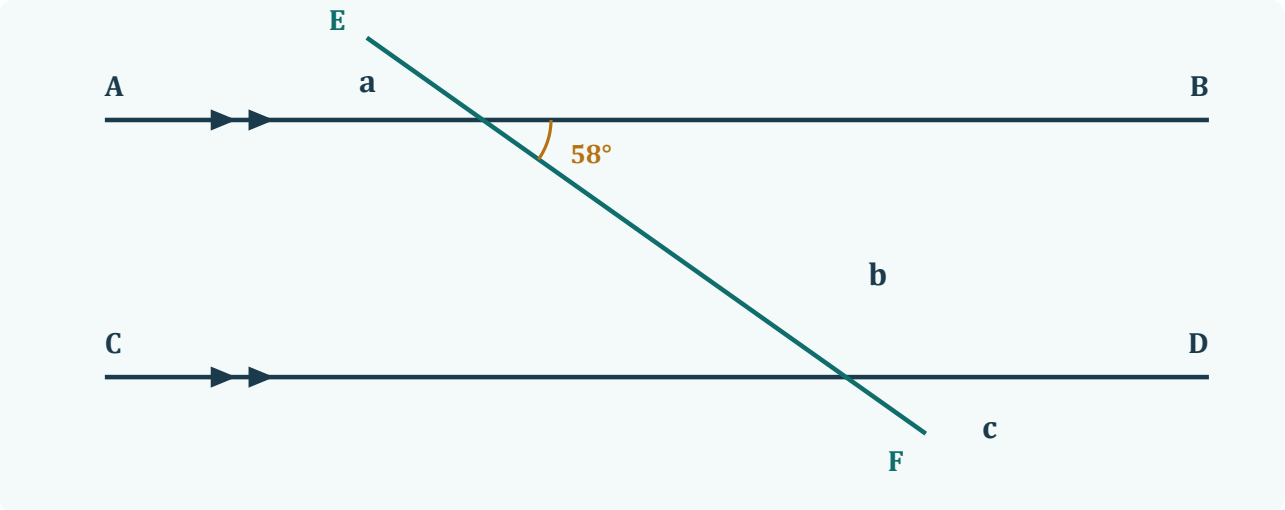
8.2 A relationship has the rule: output = −2 × input + 6. **(3)**

8.2.1 Find the output when the input is −3. (1)

8.2.2 Find the input when the output is 0. (2)

8.3 Determine the equation of the straight line passing through (0 ; −2) and (3 ; 4). **(3)**

QUESTION 9 · Geometry · 10 marks



AB is parallel to CD. EF is a transversal. The marked 58° angle lies between AB and the transversal, on the side towards B. Angle a is vertically opposite that 58° angle. Angle b is co-interior with it. Angle c corresponds to it.

9.1 In the diagram, AB ∥ CD and EF is a transversal. Determine, with reasons: **(6)**

9.1.1 the size of a, the angle vertically opposite the given 58° angle (2)

9.1.2 the size of b, the co-interior angle to the given 58° angle (2)

9.1.3 the size of c, the corresponding angle to the given 58° angle (2)

9.2 In △KLM, ∠K = 2x, ∠L = 3x and ∠M = 4x. Calculate the size of the largest angle. **(2)**

9.3 Triangle DEF is similar to triangle ABC. The vertices are written in matching order, so D matches A, E matches B and F matches C. AB = 6 cm, BC = 8 cm and AC = 10 cm. DE = 9 cm. Calculate EF. **(2)**

QUESTION 10 · Pythagoras, circles and volume · 10 marks

10.1 In a right-angled triangle the right angle lies between two sides of length 20 cm and 21 cm. Calculate the length of the hypotenuse. **(3)**

10.2 A circle has a diameter of 20 cm. Use π = 3,14. **(4)**

10.2.1 Calculate the circumference. (2)

10.2.2 Calculate the area. (2)

10.3 A closed cylinder has a radius of 7 cm and a height of 10 cm. Use π = 22/7. Calculate the volume. **(3)**

QUESTION 11 · Transformations · 4 marks

11.1 Point P(−4 ; 3) is given. Write the coordinates of the image of P after each transformation. **(4)**

11.1.1 Reflection in the y-axis. (1)
11.1.2 Reflection in the line y = x. (1)
11.1.3 A translation of 2 units right and 5 units down. (2)

OPTIONAL QUESTION

QUESTION 12 · Data and probability · 10 marks

Optional. Answer this question only if your teacher included data handling and probability. The 2025 national teaching plan uses Term 4 for transformations, patterns and revision. Platinum still teaches these chapters.

12.1 The ages, in years, of nine players are 12, 14, 14, 15, 15, 15, 16, 17 and 18. **(5)**

12.1.1 Determine the median. (1)
12.1.2 Determine the mode. (1)
12.1.3 Determine the mean. Give the exact answer, or round it to two decimal places. (2)
12.1.4 Determine the range. (1)

12.2 A spinner is divided into 8 equal sectors: 3 yellow, 3 red and 2 blue. **(5)**

12.2.1 Write P(red) as a simplified fraction. (2)
12.2.2 Write P(not yellow) as a simplified fraction. (2)
12.2.3 The spinner is spun 80 times. How many times do you expect blue? (1)

End of the question paper. Total for Questions 1 to 11: 100 marks.